

```
In [4]: import sys
import os

project_root = os.path.abspath("..")
if project_root not in sys.path:
    sys.path.insert(0, project_root)
```

```
In [5]: import numpy as np
from data.data_loader import generateGauss
from utils.integral import grid
from Models.clustering import HCF
```

```
In [ ]: bandwidth = 0.01
grid_x    = grid(bandwidth, start=-5, end=15)
mu        = np.array([0.3, 4.0, 9.1, 1.0, 5.5, 8.0, 4.8])
sig       = np.array([1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0])
F_data    = generateGauss(mu, sig, grid_x)
```

```
In [7]: cluster=HCF.Model(
    grid_x=grid_x,
    max_depth=7,
    distance_metric='L1',
    linkage='complete',
    bandwidth=bandwidth
)
cluster.fit(F_data)
cluster.print_tree()
```

```
├─ root (7 samples): [0, 1, 2, 3, 4, 5, 6]
│   ├── root0 (2 samples): [0, 3]
│   │   ├── root00 (1 samples): [0]
│   │   └── root01 (1 samples): [3]
│   └── root1 (5 samples): [1, 4, 6, 2, 5]
│       ├── root10 (3 samples): [1, 4, 6]
│       │   ├── root100 (1 samples): [1]
│       │   └── root101 (2 samples): [4, 6]
│       │       ├── root1010 (1 samples): [4]
│       │       └── root1011 (1 samples): [6]
│       └── root11 (2 samples): [2, 5]
│           ├── root110 (1 samples): [2]
│           └── root111 (1 samples): [5]
```